

Was Russell's Simple Theory of Types an *Ad Hoc* Solution to Russell's Paradox?

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Abstract

When Bertrand Russell discovered his infamous paradox in 1902, it threatened the logicist project to establish the foundations of mathematics. The logicist movement aimed to prove that the truths of mathematics are simply truths of logic by deriving mathematics from purely logical principles. However, the paradox would overturn some fundamental logicist assumptions about classes and propositional functions. Russell's proposed solution was a theory of types, known in its earlier form as the *simple theory of types* before its evolution into the *ramified theory of types*. Simple type theory excludes Russell's paradox by preventing certain sorts of self-reference which can lead to paradox, implementing a hierarchy of logical types, an approach which is criticised for being *ad hoc* and ontologically excessive. In this thesis, I explore the connection between simple type theory and Russell's lesser-known *substitutional theory* and evaluate their viability as solutions to the paradox. I demonstrate that the substitutional theory provides a philosophically coherent basis for the exclusion of Russell's paradox according to a simple theory of types. I oppose interpretations which identify simple type theory with the informal *doctrine of types* given in Appendix B of the 1903 *Principles of Mathematics*. On my view, criticisms of simple type theory are misdirected towards the doctrine of types, which was merely setting out the conditions that ought to be met by a sufficient logicist theory. I advance an interpretation of Russell's simple type theory as identical to the mature substitutional theory of 1906, within which the conditions set by the informal doctrine of types are satisfied. I conclude that Russell's proposed simple type theory, though ultimately inconsistent and insufficient to fully solve the paradoxes, was not *ad hoc* in its attempt to solve Russell's paradox.

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§1 Introduction

This thesis will explore the emergence and significance of Russell's paradox to logicism and evaluate Russell's attempt at solution within his early development of type theory. While Russell's type theory is best known in its ramified form as the foundational system of Whitehead and Russell's *Principia Mathematica* (1910), my present concerns lie with type theory prior to its ramification, developed between 1903 and 1907 and now known as *simple type theory* (henceforth, STT). In this thesis, I will refute criticism that STT is *ad hoc* or artificial by arguing for an interpretation of STT which recognises the significance of Russell's little-known *substitutional theory* to not only STT, but the development of type theory as a whole. I argue that the criticisms are misdirected at Russell's earliest exposition of type theory, which appeared in 1903's the *Principles of Mathematics*. Presented as the *doctrine of types*, it is often mistakenly interpreted as Russell's sole treatise on STT. I argue that when STT is instead identified with the mature substitutional theory of 1906, rather than the informal doctrine of types, the philosophically coherent principles upon which Russell intended to found type theory are clear.

In §2, I set out the logicist context within which the paradoxes would be so significant, especially the vital importance of classes and their derivation from propositional functions. In §3, I explain Russell's paradox in terms of both classes and propositional functions, and demonstrate its dramatic consequences for logicism in overturning the paradigmatic *naïve* conception of classes. In §4, I give an account of STT which shows how it prevents the formulation of both *extensional* and *intensional* versions of the paradox. I describe the criticisms of STT, namely its inconsistency and therefore failure to truly, completely solve Russell's paradox, as well as its supposedly *ad hoc* philosophical claims. The former is undeniable, but I oppose the latter. In §5, I refute the criticism of STT as *ad hoc* with my interpretation of Russell's STT as identical to the substitutional theory. I introduce the substitutional theory and explore the historical context contributing to how it is neglected in interpretations of type theory. I give an account of the substitutional logic to demonstrate the theory's successful, philosophically grounded exclusion of Russell's paradox, alongside its generation of

precisely the type hierarchy of STT. I conclude that Russell's proposed solution of Russell's paradox by the substitutional theory, and thus by STT, was not *ad hoc*. In §6, I conclude my argument and advocate for its wider significance.

§2 Logicism & Classes¹

The movement of *logicism*, advanced by Frege and Russell, is the doctrine that various branches of pure mathematics reduce entirely to logic. Were logicism to obtain, it would invest mathematics with all the epistemological and metaphysical advantages of logic. The truths of mathematics would be truths of logic, and the ontological status of mathematical objects, such as numbers, would not have to be assumed as mysterious, *sui generis* entities, but instead would be analysable into purely logical concepts. Logicism primarily addresses, though is not limited to, real analysis and arithmetic. In practice, logicism has two goals: to restate relevant mathematical concepts in terms of only logical concepts, and to produce a collection of logical axioms (basic sentences which are assumed to be true) from which the relevant mathematical theorems could be derived. If successful, the language and ontology of mathematics would thus be subsumed under the language and ontology of logic (Godwyn & Irvine, 2003, 171-172).

Seeking mathematical certainty and rigour, the axiomatic foundations of mathematics were studied by Weierstrass, Cantor and Dedekind in the 1800s. Cantor had developed a revolutionary mathematical theory of classes (the aggregations or collections of entities, individuated by their members or constituents) from which he derived the transfinite cardinal numbers, developing an arithmetic for strictly differing sizes of infinity (Grattan-Guinness, 2003, 53). Following this, Peano produced an axiomatisation of arithmetic which defined the class of natural numbers (the positive whole numbers, denoted $\mathbb{N}$) and their arithmetic properties. Five axioms are retained today as the *Peano axioms* of arithmetic (Van Heijenoort, 1967, 83). By first giving definitions of 0, a successor function, and formal mathematical induction, Peano inductively defines the

¹ I use the logicist term *class* to emphasise the distinction from the modern mathematical objects of sets, which are constructed within Zermelo-Fraenkel axiomatic set theory (ZFC).

class $\mathbb{N}$ as the collection satisfying: (1) 0 is a member of $\mathbb{N}$ and (2) for any n in $\mathbb{N}$, its successor, $n + 1$, is also in $\mathbb{N}$. Arithmetic operations of multiplication, addition, subtraction, and division are all further definable in terms of the successor function (Peano, 1889, 94-97).

Peano's axioms of arithmetic set an example for the first logicians, with both Russell (1903) and Frege independently producing works on logicism. Frege had informally set out his logicist intentions in the *Foundations of Arithmetic* (1884), later formalising these ideas into a logical axiomatisation of arithmetic in the *Basic Laws of Arithmetic* (1893). Frege, dissatisfied with the justification for some of Peano's primitive assumptions, introduced a relation of *equinumeracy*, which holds between classes with the same cardinality (the same number of members). Equinumeracy is an equivalence relation of classes, generalising the notion of equality to class 'size'. The number n is then defined as the equivalence class of all n -membered classes. The number 1 is therefore the class of all single-membered classes, 2 the class of all pairs, 3 the class of all triples, and so on. Equinumeracy between two classes is defined without reference to numbers, but instead by the existence of a bijection (1-1 correspondence) between their members. Therefore, the equivalence classes of equinumeracy are constructed prior to any assumption of numbers, and any apparent circularity in the definition is avoided (1893, 96-97). With the addition of his own class-theoretic definitions of 0 and the successor function, Frege followed Peano in giving his own logical axiomatisation of arithmetic. Frege's axiomatisation of numbers and arithmetic in terms of (supposedly) logical concepts (classes, relations, bijection, etc.) exemplifies the logicist reduction of the mathematical to the logical.

The notion of a *class* and the relation of class membership is therefore indispensable to Frege's logicism. According to most early logicist theories, classes are aggregates or collections of objects which can be considered objects in their own right. Classes may be defined *extensionally*, by listing their members one by one, or *intensionally*, by providing a necessary and sufficient condition for class membership, called a *class concept* by Frege (for Russell, a *propositional function*). Therefore, it is often convenient to speak of an intensional class concept or propositional function, denoted $\phi\hat{x}$,

alongside its corresponding extension, denoted $\{x: \phi x\}$, the class of all entities x such that ϕx is a true proposition.² Intensional definition is essential to any theory of mathematics which employs classes, for the infinities studied by mathematics can never be defined extensionally in our finite lifetimes. While propositional functions and classes differ in intensionality (different propositional functions may define the same class), it is important that what may be said about one is often applicable to the other.

For my discussion, Frege's most relevant axiom is his *Basic Law V*, the axiom of *unrestricted comprehension* which governs the intensional definition of classes from class concepts. It states that two classes are the same iff their class concepts are satisfied by precisely the same arguments (1893, 72).³ *Basic Law V* makes the apparently innocuous assumption that every predicate is a class concept with a well-defined extension. That is, that given any condition, there exists a class of objects satisfying that condition. This assumption would be overturned by Russell's paradox, implicating theories which included an axiom of unrestricted comprehension, now known as *naïve* theories of classes.

§3 Russell's Paradox

Russell had independently developed his own treatise on logicism, the *Principles of Mathematics* (1903), with many similarities to Frege. Where Frege had class concepts and equinumeracy, Russell employed propositional functions and the *similarity* relation of classes. Just before the *Principles* was published, Russell wrote to Frege about his discovery of a contradiction which followed from Basic Law V. A logical system *must* be consistent due to the explosion principle: if a single contradiction is provable, then *anything* is provable within that system, and the system's integrity collapses. The severity of the contradiction is apparent in Frege's reply, that the contradiction "has shaken the basis on which I intended to build arithmetic" (Frege, 1902, 127).

² My notation throughout is mostly due to Russell, but may be modernised for clarity.

³ In modern notation, the law states: $[\{x: \phi(x)\} = \{x: \psi(x)\}] \equiv (x)[\phi(x) \equiv \psi(x)]$

Recalling that the early logicians' naïve classes may contain other classes, as in the case of Frege's natural numbers, classes may or may not be contained by themselves. For example, the class of men is not a member of itself, being not a man, but the class of non-men *is* a member of itself, being also not a man. The extensional form of Russell's paradox proceeds: consider the condition '*is not a member of itself*'. If, by Basic Law V, we assume this condition corresponds to a class concept, we obtain the Russell class, $C = \{X: X \notin X\}$,⁴ the class of all classes which are not members of themselves. A contradiction is simply derived by considering whether the Russell class, C , contains itself. If $C \notin C$, it satisfies its own condition for membership, therefore $C \in C$, and vice versa. We obtain the class form of Russell's paradox: $C \notin C \equiv C \in C$. Other mathematical contradictions had previously arisen, but they often concerned mathematical notions (e.g. the Burali-Forti paradox⁵) and, one hoped, could be overcome by local adjustment of theory. However, Russell's paradox concerned logicism's most fundamental assumptions.

In fact, Russell had originally posed the paradox to Frege in terms of predicates rather than classes:

Let w be the predicate: to be a predicate that cannot be predicated of itself. Can w be predicated of itself? From each answer, its opposite follows" (Russell, 1902, 125)

The paradox, in this wording, was resisted by a facet of Frege's philosophy that was to inform Russell's later attempts at solution. Frege held an ontological distinction between *concept* and *object*, originally motivated by the *unity of the proposition*, a puzzle to both Russell and Frege. In brief, propositions seemingly possess a unity that does not survive analysis into their basic constituents. The analysis of "Socrates is mortal" into "Socrates" and "Mortality" could just as easily be reconstructed into "Mortality is Socrates" (Russell, 1903, 49). According to Frege, the propositional structure is

⁴ I write $a \in A$ to mean " a is a member of A " and may equivalently say " A contains a ".

⁵ See Russell (1903, 323).

preserved by the distinct ontological status of “Socrates” as an object and “Mortality” as an unsaturated concept (1892, 54). The unsaturated concept requires supplementation by an object as argument, analogous to how an ‘unsaturated’ mathematical function becomes ‘complete’ when supplied with an argument (1891, 24). This distinction provided Frege with a temporary solution to this form of the paradox: since a predicate is a first level function that may only take an object as an argument, a predicate cannot be predicated of itself (1892, 50). However, this was unsatisfactory to Russell. Firstly, the ontological stratifications of Frege’s distinction between concept and object were unpalatable to Russell, who maintained that every logical subject be of the same logical type (Stevens, 2003, 225). Secondly, Russell found the paradox could be reworded to apply intensionally to class concepts (or propositional functions):

A class-concept may or may not be a term of its own extension. “Class-concept which is not a term of its own extension” appears to be a class-concept. But if it is a term of its own extension, it is a class-concept which is not a term of its own extension, and vice versa. Thus, we must conclude, against appearances, that “class-concept which is not a term of its own extension” is not a class-concept.
(1903, 102)

The two explanations are similar, but it is their subtle difference which allows the latter to circumvent Frege’s concept/object distinction. Russell explicitly proves the existence of a predicate which, on pain of contradiction, cannot define a class. In other words, the inconsistency overturns the assumption of *Basic Law V*, that every predicate corresponds to a class-defining property (Quine, 1961, 13). The discovery of the paradox caused the rejection of naïve theories of classes from mathematics and logicism alike, leaving a vacuum for a consistent axiomatisation of classes upon which to establish logicism.

§4 Russell's Proposed Solution

§4.1 The Doctrine of Types & Simple Type Theory

The paradox showed that a theory of classes could not adopt an axiom of unrestricted comprehension; limitations had to be imposed on which predicates (mere bits of language) could be propositional functions, and so which predicates could define classes. Thus began Russell's extensive project to solve the paradoxes within logicism, culminating in the type theory of Russell and Whitehead's *Principia Mathematica* (1910). The theory began with Russell's prototypical *doctrine of types*, which he hoped to be "a first step towards the truth" (1903, 523), in which the informal principles of a theory of types are laid out. The early doctrine of types is anachronistically known as *simple type theory* (STT), prior to its later evolution into *ramified type theory* (RTT), first appearing in *Mathematical Logic as Based on the Theory of Types* (1908), before its later appearance in *Principia*.

Russell's evolving characterisation of propositions and propositional functions is vital to understanding type theory. While *Principia* would rest on Russell's *multiple relation theory of judgement*, which has no ontological commitment to propositions, the *Principles* adopted an ontology of *Russellian propositions*, as they are now known. A Russellian proposition is a mind-independent, structured, complex entity constituted of the entities it is 'about'. For example, the proposition 'Socrates is mortal' is a complex object constituted by the entities *Socrates* and *Mortality*. The proposition is true iff such entities really exist which have the asserted property or stand in the asserted relation to each other. The logicist movement generally adopted a Platonist ontology (Quine, 1948, 84), clearly apparent in Russell's ontology in the *Principles*, in which the proposition 'Socrates is mortal' and the property of mortality are as much existing entities as Socrates himself, albeit as universals rather than particulars. While decidedly not set theoretic entities, Russellian propositions can be represented as ordered sets containing their constituents, as in $\langle \text{Socrates}, \text{Mortality} \rangle$ (Stevens, 2010, 118).⁶

⁶ The source of a proposition's structure, represented here by set-theoretic ordering, is precisely the problem of the unity of the proposition.

Furthermore, analogous to Frege's concepts, Russell spoke of *propositional functions*. A propositional function $\phi\hat{x}$ (or $\phi(\hat{x}_1, \dots, \hat{x}_n)$ for propositional functions of n arguments) is a statement 'about' an undetermined x and which becomes a proposition when x is given a fixed determination, for example, $\hat{x}$ is mortal. The circumflex notation in $\phi\hat{x}$ distinguishes the propositional function itself from an ambiguous value of the function, as in ϕx . When x is given the definite determination a , ϕa is an unambiguous value of $\phi\hat{x}$ (Whitehead & Russell, 1910, 14-15).

Russell's early doctrine of types suggested that propositional functions should be of a different logical type to their possible arguments, in the likeness of Frege's concepts which could only be significantly predicated of objects (though Frege's ontological type distinctions remained unpalatable to Russell). The doctrine of types posits that for any propositional function $\phi\hat{x}$, there is a *range of significance* containing the entities which result in a proposition when supplied as argument to $\phi\hat{x}$. These ranges are sums of *types*, defined such that if a function is significant for an argument of type n , that function must be significant for any other argument of type n . The type hierarchy begins with the most basic type of individuals, type 0. Then, type 1 contains propositional functions which are significant for arguments of type 0 (individuals). In general, a propositional function of type n , denoted by the type-indexed $\phi^n\hat{x}$, is only possibly significant for an argument of at most type $n - 1$. Thus $\phi^{n+1}x^n$ is significant, but $\phi^n x^m$ is ill-formed and meaningless unless $n > m$. By restricting the possible significant arguments of propositional functions to only certain types, the classes they define are similarly restricted. The types of propositional functions thus have their extensional reflection in a hierarchy of types of classes: individuals, classes of individuals, classes of classes, and so on (1903, 523-525).⁷

It appears that restricting possible arguments to certain types bluntly declares the paradoxical formulation meaningless. The intensional version of the paradox concerns

⁷ The hierarchy is yet more expansive when propositional functions of multiple arguments and their corresponding relations-in-extension (classes of n -tuples) are accounted for.

a propositional function of type n , ‘ $\hat{x}$ is not a term of its own extension’, written $\phi^n \hat{x}$. Then the type of its argument must be at most $n - 1$, which the required paradoxical formulation, $\phi^n(\phi^n \hat{x})$, clearly violates. Thus, STT excludes $\phi^n(\phi^n \hat{x})$ as an ill-formed, meaningless expression (insignificance is additionally inherited by the extensional formulation, $X^n \in X^n$) and Russell’s paradox is apparently solved. The doctrine of types and its solution to the contradiction are presented rudimentarily and left unformalized in Appendix B, with Russell acknowledging “it requires, in all probability, to be transformed into some subtler shape” (1903, 523). I will further explore this point in §5 and, for now, urge against identifying this account of STT with Russell’s pre-ramification intentions for a formal theory of types.⁸

§4.2 The Criticisms of STT

STT, when identified with this prototypical doctrine of types, is criticised for being *ad hoc* and artificial for its unfounded declaration of the paradox as meaningless, and intolerable imposition of ontological type stratifications. Quine says, “Russell sees the universe as dividing into levels, or *types*” (1967, 151). To simply declare the paradox as meaningless, dividing up the universe to make it so, is indeed an unacceptable solution. To posit this for the sole purpose of avoiding paradox would be crude and artificial. Russell went to great lengths in the *Principles* to avoid admission of any ontological type distinction between entities, adopting his *doctrine of the unrestricted variable*: “any conceivable entity may be substituted for any one of our variables without impairing the truth of our proposition” (1903, 7). To restrict a variable’s domain to only certain types of entity is to be committed to such ontological divisions. STT, in the rudimentary form of the doctrine of types, certainly appears incompatible with unrestricted variation.

I hold that whether STT is an *ad hoc* attempt at solution should be established by (1) whether the hierarchy of types is ontological, apparent by its compatibility with unrestricted variation and (2) whether the exclusion of Russell’s paradox is justified

⁸ This presentation illustrates the traditional interpretation of Russell’s intent for STT, borrowing notation from Russell’s later formulations of RTT and incorporating the type-indexing of modern formalisations of STT.

within the system, or merely artificially imposed. I will show that Russell was able to reconcile the doctrine of types with unrestricted variation in his little-known *substitutional theory*, within which the paradox's lack of meaning is a product of the basic substitutional grammar. I thus refute the criticism that Russell's attempt to solve the paradoxes within STT was *ad hoc*.

It must be noted that, no matter its philosophical coherence, simple type restrictions are insufficient to exclude all the various paradoxes. Whether a simple type hierarchy is realised within the traditional formulation or the substitutional theory, further inconsistencies are derivable, such as the paradoxes of *Epimenides* or the *Liar*. STT alone is inconsistent, and thus cannot establish logicism. These further paradoxes would lead Russell to ramify type theory, introducing an additional propositional hierarchy of orders that would draw yet more criticisms of artificiality. RTT is ultimately judged to have failed to establish logicism (Urquhart, 2003, 303-304). However, I argue for only the philosophical coherence of Russell's STT prior to ramification. My argument is thus exclusively aimed at refuting the criticism that STT was an *ad hoc* solution to Russell's paradox.

§5 The Substitutional Theory & STT

I will defend STT against *ad hoc* criticisms by arguing for two key ideas. The first is that Russell's substitutional theory (1906b), first known as the *no classes* theory (1906a), successfully prevents the formulation of Russell's paradox by incorporating types without abandoning unrestricted variation, and thus is not *ad hoc* in its attempt to solve Russell's paradox. I will show that the substitutional theory's type hierarchy is plainly not ontological, but applies only to what I will elucidate as the linguistically convenient fictions that are the substitutional theory's classes and propositional functions. Russell's paradox is not *declared* meaningless within substitution but is simply impossible to formulate within the natural substitutional grammar.

My second point addresses what ought to be identified as *Russell's STT*, i.e. Russell's most sophisticated formulation of an unramified theory of types. I hold that the

formalised system of STT which ought to be attributed to Russell is the substitutional theory. Russell's 1903 doctrine of types is often assumed to be Russell's sole and informal account of STT, and is the primary target of *ad hoc* criticism.⁹ My interpretation is that the doctrine of types was not a mature theory, but was intended as an exposition of the guiding principles that ought to be met by any theory hoping to offer a satisfactory solution to the paradox. Accordingly, Russell developed the substitutional theory to provide the necessary philosophical justification for implementing the doctrine of types. From this, it follows that Russell's proposed solution of his paradox by STT may claim all the philosophical virtues of the substitutional theory (which will be expounded shortly), and thus was not *ad hoc*.

I therefore oppose a range of critical interpretations of how the 1903 doctrine of types and the 1906 substitutional theory bear on interpretations of Russell's STT. On the traditional interpretation, such as Quine's, Russell abandoned his 1903 type theory to explore alternative avenues to solving the paradox, including the no classes (substitutional) theory, within which "the theory of types is not even mentioned" (1967, 150).¹⁰ On this view, STT is often identified with anachronistic formalisations which impose type-indexing onto its variables, somewhat resembling an 'unramified' version of Russell's RTT.¹¹ My own presentation of STT above is closer to these modern formalisations with its type-indexed variables, and is presented as such both to illustrate the source of the *ad hoc* criticism, as well as to contrast with the forthcoming exposition of the substitutional theory.

Historically, the significance of the substitutional theory to the development of type theory has been overlooked. Russell gives the main account of the theory in *On the Substitutional Theory of Classes and Relations* (1906b), and the earliest published exploration of ramification (based in substitution) is given in *On 'Insolubilia' and their*

⁹ The SEP entry on Russell's Paradox gives the traditional interpretation: "Russell's type theory thus appears in two versions: the "simple theory" of 1903 and the "ramified theory" of 1908. Both versions have been criticised for being too *ad hoc* to eliminate the paradox successfully." (Irvine & Deutsch, 2021). See also Urquhart (2003, 289-290).

¹⁰ See also Quine (1940, 163-164) & (1966, 76-77).

¹¹ See Church (1984) and Copi (1971) for such formalisations.

Solution by Symbolic Logic (1906c). These two key papers did not appear in English until 1973, the former withdrawn from publication and the latter published only in French. Further vital manuscripts containing a formal axiomatisation of substitution and further initial explorations of ramification within the substitutional theory (*On Substitution* (1906d) & *The Paradox of the Liar* (1906e)) only became widely available in 2014. The substitutional theory has gained recognition in the work of Landini (1998, 2003), but (perhaps due to its abandonment by the time of *Principia*) it is neglected in most accounts of type theory, and its consequences remain not widely recognised.

§5.1 The Substitutional Grammar

Turning to the substitutional theory itself, I will give a brief account of the logic itself to show how the substitutional grammar prevents formulation of Russell's paradox, while providing a theory of 'classes' from which Russell intended to derive mathematics. I will further demonstrate how the substitutional grammar eliminates ontological commitment to both the classes and propositional functions which STT stratifies into types. As I will demonstrate, the type hierarchy is therefore not ontological, but a fictional by-product of the substitutional grammar.

The implementation of a non-ontological type hierarchy is realised within the substitutional theory's adherence to unrestricted variation. The substitutional theory's variables stand unrestricted for any of the totality of entities in one's ontology. The ontology of the substitutional theory is one of Russellian propositions and their constituents, together with the primitive operation of substitution whereby an entity occurring within a proposition is substituted for another. For example, the entity *Plato* may be substituted for every occurrence of the entity *Socrates* within the proposition *Socrates is mortal*; this results in the proposition *Plato is mortal*. Within a set theoretic representation of Russellian propositions, this substitution may be understood as the operation which bears $\langle \text{Socrates}, \text{mortality} \rangle$ to $\langle \text{Plato}, \text{mortality} \rangle$. As will be further demonstrated, there is no ontological type distinction between a proposition and its constituents, all are entities of the same sort. Russell also takes implication as primitive, represented by the logical constant, $\supset$, which gives a well-formed formula

(wff) when flanked by *terms*, as in $a \supset b$ (different from the modern logical connective $\rightarrow$ which gives a wff when flanked by wffs), again towards maintaining unrestricted variation such that a and b need not have any special ontological status for $a \supset b$ to be significant.

Where p , a , x , and q stand for any entities, the fundamental statements of substitution are defined: $p/a; x! q$ is a proposition, meaning ‘ q results from substituting x for every occurrence of a in p ’. Within this statement, p/a is called a *matrix*, for which p is the *prototype* and a is the *argument*, where a may or may not be a constituent of p . Importantly, while the prototype and argument each stand for single entities, a matrix does not. Furthermore, $p/a; x$ means ‘*the* q for which $p/a; x! q$ holds’, i.e. the entity that results from replacing all occurrences of a in p with x .

Substitution alludes to the idea of similarity of structure; p and q are alike in every sense, except where a appears in p , x appears in q . Thus $p/a; x$ names an entity with a certain structural similarity to p . While p and q may be any entities, it is convenient to think of them as mostly standing for propositions, likewise a and x for logical subjects. While this assists with intuition, it is vital to the substitutional theory’s adherence to unrestricted variation that this is not necessary. Neither propositions nor logical subjects demand any special ontological status or type to be meaningfully spoken about in the language of substitution.

Just as the substitution of single entities corresponds to the notion of propositional functions of one argument, Russell meticulously defines simultaneous multiple substitutions to emulate propositional functions of multiple arguments. The statement $p/(a, b); (x, y)! q$ roughly means “ q results from replacing all (if any) occurrences of a in p with x and b in p with y ”. This is a simplified definition, as much care must be taken with these highly technical definitions to avoid fallacious instances of substitution, for example in the case of attempting to substitute an entity a into a p which already contains a as a constituent; similar opportunities for fallacy abound in multiple substitution. However, Russell’s definition successfully allows substitution to operate

as a structure preserving operation, whether of a single entity or multiple entities. The matrices p/a and $p/(a, b)$ do not stand for entities, but there is a clear difference in grammatical form; we will see that this difference corresponds to a non-ontological type distinction.

§5.2 Substitutional Types, Classes & Logicism

The most radical advancement of the substitutional theory was the elimination of classes as entities (hence it was originally known as the *no classes* theory). Equipped with his new *theory of descriptions* (1905), Russell's approach was to deny classes any independent ontological status, but still facilitate their use in mathematics by providing contextual definitions for class symbols. That is, class talk is a mere *façon de parler* and "statements apparently about a class will only be significant when they can be analysed into statements about all or some of the members" (1906b, 166). While a class symbol has no independent meaning (it is an *incomplete symbol*), every significant proposition in which a class symbol appears can be assigned a meaning which says nothing at all about classes. Thus, classes may be countenanced by a logical foundation for mathematics without admitting their existence as entities, and thus without ontologising type distinctions.

Russell's ontological elimination of classes and their treatment as incomplete symbols is implemented by the association of classes with matrices, such as p/a or $p/(a, b)$. A matrix is not an entity by itself, but is instead defined by the entities that make up its prototype and argument(s). It is a "mere operation like d/dx [...] wholly devoid of meaning [and only] significant as parts of appropriate propositions" (1906b, 170). To facilitate the employment of classes in deriving mathematics, contextual definitions are given for the use of class symbols:

$$\begin{aligned} (1) \ x \in p/a &= (\exists q)(p/a; x! q \ \& \ q) && \text{Df.}^{12} \\ (2) \ (x, y) \in p/(a, b) &= (\exists q)(p/(a, b); (x, y)! q \ \& \ q) && \text{Df.} \end{aligned}$$

¹² The notation " $\dots = \dots$ Df." establishes definitional equivalence between the definiendum and definiens which flank " $=$ ".

(1) defines the assertion of class membership: x is a member of p/a iff there is a true proposition which results from the substitution of x for a in p . (2) similarly asserts a couple's membership in a dyadic relation-in-extension, or class of couples. Two matrices are said to be equivalent, and thus the same class (written $p/a = s/t$), iff for every x , $p/a; x$ and $s/t; x$ are both true or both false (1906b, 173). Thus the substitutional theory may assert equality between classes in the desired extensional sense: two classes are equal if they contain precisely the same members. Matrices can represent extensional classes but have the intensionality of propositional functions, as extensionally equivalent matrices are still distinguishable: p/a may define the same class as s/t , and thus be considered equivalent, but the symbols may still be distinguished if $p \neq s$ or $a \neq t$.

It is with the contextual definition of classes of classes, defined as special cases of double substitutions, that substitutional types begin to emerge:

$$(3) \ r/c \in q/(p, a) = (\exists t)(q/(p, a); (r, c)! t \ \& \ t) \quad \text{Df.}$$

(3) asserts a class's membership in a class of classes, provided the matrix satisfies certain conditions to ensure the extensional equivalence required of classes. "The matrix $q/(p, a)$ is called a *class of classes* if, for all values of r, c, r', c' , provided $r/c = r'/c'$, then $q/(p, a); (r, c) [=] q/(p, a); (r', c')$ " (1906b, 176). Satisfying this, the definiens of (3) may be equivalently written $r/c \in q/(p/a)$. The definienda of (2) and (3) clearly have identical logical form, thus this condition defines a merely grammatical difference between substitutional classes of classes and dyadic relations-in-extension.

Similar contextual definitions define classes of classes of classes as special matrices of triple substitutions, and so on. With a coherent notion of class membership and the resulting hierarchy of classes (if only of mere symbolic fictions), the axioms of arithmetic may be given in the style of Frege and Peano. Russell explicitly demonstrates the construction of the cardinal and ordinal numbers within the substitutional theory,

up to the transfinite ω_n and $\aleph_n$ for finite values of n , each corresponding to classes with logical type n (1906b, 183).

The substitutional theory's type hierarchy is made explicit by Russell: "A matrix of the form p/a is called of the first type; one of the form $p/(a, b)$ is called of the second type, and so on." (1906b, 176-177). All the prescriptions of the doctrine of types are incorporated into the substitutional grammar. Returning to Russell's paradox, we see that the paradox requires a significant substitutional statement, $p/a \in p/a$, to emulate $C \in C$. Within the language of substitution, this would require $p/a; [p/a]$ to name an entity, meaning 'the result of replacing a in p by [the result of replacing a in p by ...]'. The statement is clearly nonsense and impossible to significantly formulate in the language of substitution, entailed by the incompleteness of class symbols, their mention meaningless unless placed in an appropriate context. Thus, Russell's paradox is meaningless within substitution due its grammar, rather than as an *ad hoc* imposition.

Russell additionally notes the theory's "drastic pedantry" (1906b, 189) to the doctrine of the unrestricted variable: to be an entity is to be capable of being a term, or logical subject. Every variable in the basic formulae $(p/a; x! q)$ of the substitutional theory is of the same kind and may stand for any entity. Although in the majority of relevant cases p stands for a proposition, and a for an entity that occurs free in p as logical subject, the language does not require these variables to be of certain types. Despite this, the doctrine of types appears as a natural consequence of the grammatical structure of the logic, preventing the formulation of Russell's paradox while still allowing the derivation of mathematics.

§5.3 Ramification

While I have shown that the substitutional theory was not *ad hoc* in its solution of Russell's paradox, it remains an inconsistent system without ramification. The problem, foreseen by Russell, was that "some contradiction should be found to result from the assumption that *propositions* are entities" (1906b, 188). That contradiction, concerning quantifying propositions, was shortly found, referred to by Landini as Russell's " p_0/a_0

paradox”(2003, 272). In the manuscript *On Substitution*, Russell’s diagnosis is given: “ $(x). \phi x$, it would seem, must be inadmissible if $(x). \phi x$ is one of the values of x , so that it implies $\phi\{(x). \phi x\}$ [...] This wants a hierarchy of propositions” (1906d, 138).

In fact, Russell initially explored ramification within the substitutional theory (1906c, 1906d, 1906e). In *Principia*’s predecessor, *Mathematical Logic as Based on the Theory of Types*, Russell expounds a form of RTT which imposes a hierarchy of orders of propositions, but subtly retains a substitutional basis for types (1908, 238). It is clear that Russell intended the substitutional theory to be STT. However, ramification undermines the advantages provided by the substitutional theory, which is not viable without a full ontology of Russellian propositions, but inconsistent without ramification (Stevens, 2005, 63).

S6 Conclusion

I have demonstrated the significance of Russell’s paradox to the logicist movement and explored Russell’s attempted solution by simple type theory. Despite STT’s inconsistency, and therefore failure to completely solve Russell’s paradox, I have argued that criticisms that it was an *ad hoc* attempt at solution are misplaced. While Russell’s earliest theory of types appears to be *ad hoc*, this is simply due to the mistaken attribution of STT to the 1903 doctrine of types. When Russell’s STT is instead understood to be the 1906 substitutional theory, the justification for the paradox’s lack of meaning within the system and the ensuing hierarchy of types can clearly be found in the substitutional grammar. While STT is traditionally interpreted to artificially impose type stratifications and prohibitions of paradoxical statements, we can see that the substitutional grammar resolves these issues before the first substitutional axiom is even given. Were it not for the substitutional theory’s self-defeating reliance on a Russellian ontology of propositions, it would have provided Russell with a philosophically coherent foundation for the establishment of logicism. While STT does not completely solve Russell’s paradox, it was decidedly not an *ad hoc* attempt.

The encoding of types within the substitutional theory demonstrates Russell's dedication to avoiding imposing type or order hierarchies onto his ontology. He consistently favoured abandoning ontological commitment to the entities that seemed to require stratification before accepting their division. Appreciation of this within the development of STT has important implications for Russell's following logicism and wider philosophy, perhaps most immediately to interpreting the RTT of *Principia Mathematica*.

Final Mark: 85

Saleh Zarepour

The dissertation offers an outstanding engagement with a highly technical issue about Russell's type-theoretical solution to Russell's paradox. It is very well structured and easy to follow despite the fact that the topic itself is complicated. No irrelevant discussion can be found in the essay. The author demonstrates excellent knowledge of Russell's (philosophy of) logic. I really enjoyed reading this dissertation.

It is very hard to detect a shortcoming in the essay. Nevertheless, it seems that the discussion of sections 5.2 and 5.3 could have been developed more patiently.

Graham Stevens

This is a really outstanding essay. The level of technical understanding, coupled with extensive reading of very difficult material and a genuine insight into the philosophical sophistication of Russell's logic is quite remarkable to see in an undergraduate dissertation. Your understanding of the substitutional theory is really impressive. You also do a fantastic job of grasping the general landscape of logicism in a way that is both mathematically and philosophically informed. The care and attention you have invested into getting to grips with Russell's philosophical project is superb. I really liked the way you took the Appendix B doctrine of types as setting out a formally adequate condition of a theory of types, rather than a theory in itself. This has a lot of textual support (more, in fact, than you show here although it would be unfair to criticize you for that). The way that you narrow down the focus of the essay rather than allowing yourself to be stranded

in the wider debates about logicism is excellent. Your understanding of the secondary literature and positioning of your view in relation to it is exemplary.

It is very hard to find fault with this essay, but my one complaint would perhaps be that just a tiny bit more explanation at the end as to why substitution could not survive ramification would have completed the story perfectly. I realise this may have been the inevitable consequence of the space requirements, but it felt a little abrupt.

References

Copi, I. M. 1971. *The Theory of Logical Types*. London: Routledge & Kegan Paul.

Frege, G. 1884. *Foundations of Arithmetic*, translated by J. L. Austin. 2nd revised edition (1953). Oxford: Blackwell.

Frege, G. 1891. 'Function and Concept', translated by P. Geach, in P. Geach & M. Black (eds. and trans.) 1980, 21-41.

Frege, G. 1892. 'Concept and Object', translated by P. Geach, in P. Geach & M. Black (eds. and trans.) 1980, 42-55.

Frege, G. 1893. *Basic Laws of Arithmetic*, translated and edited by M. Furth (1964). Berkeley: UCP.

Frege, G. 1902. 'Letter to Russell', in J. Van Heijenoort (ed.) 1967, 126-127.

Geach, P. & Black, M. (eds. and trans.), 1980. *Translations from the Philosophical Writings of Gottlob Frege*. 3rd edition. Oxford: Blackwell.

Godwyn, M. & Irvine, A. D. 2003. 'Bertrand Russell's Logicism', in N. Griffin (ed.) 2003, 171-201.

Grattan-Guinness, I. 2003. 'Mathematics in and behind Russell's Logicism, and Its Reception', in N. Griffin (ed.) 2003, 51-83.

Griffin, N. (ed.) 2003. *The Cambridge Companion to Bertrand Russell*. Cambridge: CUP.

Irvine, A. D. & Deutsch, H. 2021. 'Russell's Paradox', *The Stanford Encyclopaedia of Philosophy* (Spring 2021 edition), edited by E. N. Zalta.

URL = <<https://plato.stanford.edu/archives/spr2021/entries/russell-paradox/>>.

Landini, G. 1998. *Russell's Hidden Substitutional Theory*. New York: OUP.

Landini, G. 2003. 'Russell's Substitutional Theory', in N. Griffin (ed.) 2003, 241-285.

Peano, G. 1889. 'The Principles of Arithmetic', translated by J. Van Heijenoort, in J. Van Heijenoort (ed.) 1967, 85-97.

Quine, W. V. 1948. 'On What There Is', in D. H. Mellor & A. Oliver (eds.), *Properties*. New York: OUP, 74-88.

Quine, W. V. 1940. *Mathematical Logic*. Page references are to the revised edition (1951). Cambridge, MA: HUP.

Quine, W. V. 1961. 'The Ways of Paradox', in W. V. Quine, *The Ways of Paradox and other essays*. Cambridge, MA: HUP, 1-18.

Quine, W. V. 1966. 'Russell's Ontological Development', in W. V. Quine, *Theories and Things*. Cambridge, MA: HUP, 73-85.

Quine, W. V. 1967. 'Introduction to Russell [1908]', in J. Van Heijenoort (ed.) 1967, 150-152.

Russell, B. 1902. 'Letter to Frege', in J. Van Heijenoort (ed.) 1967, 124-125.

Russell, B. 1903. *Principles of Mathematics*. 2nd Edition (1937). London: Allen & Unwin.

Russell, B. 1905. 'On Denoting', in B. Russell 1973, 103-119.

Russell, B. 1906a. 'On Some Difficulties in the Theory of Transfinite Numbers and Order Types', in B. Russell 1973, 135-164.

Russell, B. 1906b. 'On the Substitutional Theory of Classes and Relations', in B. Russell 1973, 165-189.

Russell, B. 1906c. 'On 'Insolubilia' and their Solution by Symbolic Logic', in B. Russell 1973, 190-214.

Russell, B. 1906d. 'Two Drafts on Substitution', in B. Russell 2014, 126-235.

Russell, B. 1906e. 'The Paradox of the Liar', in B. Russell 2014, 317-374.

Russell, B. 1908. 'Mathematical Logic as Based on the Theory of Types', in J. Van Heijenoort (ed.) 1967, 150-182.

Russell, B. 1973. *Essays in Analysis*, edited by D. Lackey. London: Allen & Unwin.

Russell, B. 2014. *The Collected Papers of Bertrand Russell, Volume 5: Toward Principia Mathematica 1905-1908*, edited by G. Moore. Routledge.

Stevens, G. 2003. 'The truth and nothing but the truth, yet never the whole truth: Frege, Russell and the analysis of unities', *History and Philosophy of Logic*, 24: 221-240.

Stevens, G. 2005. *The Russellian Origins of Analytical Philosophy*. London: Routledge.

Stevens, G. 2010. 'Russell's Ontological Development Reconsidered', *British Journal for the History of Philosophy*, 18: 113-137.

Urquhart, A. 2003. 'The Theory of Types', in N. Griffin (ed.) 2003, 241-285.

Van Heijenoort, J. (ed.) 1967. *From Frege to Gödel: A source book in mathematical logic, 1979-1931*. Cambridge, MA: HUP.

Whitehead, A. N. & Russell, B. 1910. *Principia Mathematica*. Page references are to *Principia Mathematica* to *56, 2nd edition (1997). Cambridge: CUP.